

What is TCP/IP?

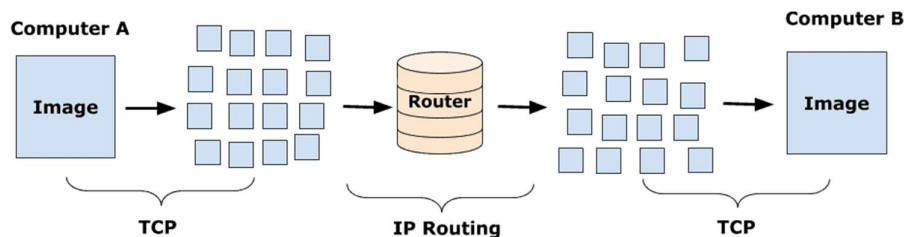
1. TCP/IP stands for Transmission Control Protocol/Internet Protocol and is a suite of communication protocols used to interconnect network devices on the internet.
2. TCP/IP is also used as a communications protocol in a private computer network (an intranet or extranet).

Port numbers range from 0 to 65535

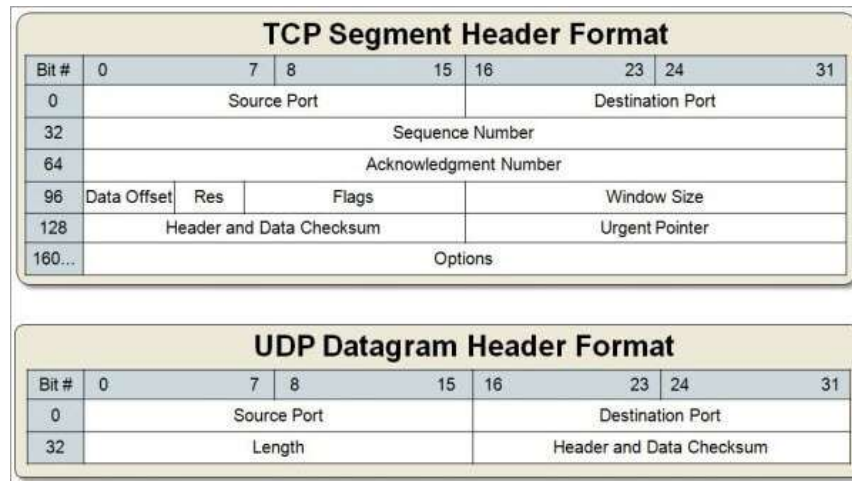
- Well-known ports range from 0 through 1023.
- Registered ports are 1024 to 49151.
- Dynamic ports (also called private ports) are 49152 to 65535.
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How IP Routing Works – Understanding TCP/IP:

1. TCP/IP, also known as Internet Protocol Suite, is a collection of communication protocols that use the internet to send and receive data.
2. A protocol is a set of rules that computers use to communicate with one another, and internet communication is founded on the **Transmission Control Protocol (TCP)** and **Internet Protocol (IP)**.
3. Each perform a different step in the process of transferring data. TCP prepares the data by breaking it down into smaller pieces of information called “packets” while also defining and maintaining the network connection.
4. IP deals with the sending and delivering of the packets from one computer or server to another using a web of routers in a process known as **IP Routing**.
5. Once the packets reach their destination, it is the job of TCP to reform them into their original state.



1. **Transmission Control Protocol (TCP):** Port Number: 6
TCP is a popular communication protocol which is used for communicating over a network. It divides any message into series of packets that are sent from source to destination and there it gets reassembled at the destination.



2. **Internet Protocol (IP):** IP is designed explicitly as addressing protocol. It is mostly used with TCP. The IP addresses in packets help in routing them through different nodes in a network until it reaches the destination system. TCP/IP is the most popular protocol connecting the networks.

3. **User Datagram Protocol (UDP):** Port Number: 17

1. User datagram protocol (UDP) operates on top of the Internet Protocol (IP) to transmit datagrams over a network.
2. UDP does not require the source and destination to establish a three-way handshake before transmission takes place.
3. Additionally, there is no need for an end-to-end connection.
4. Since UDP avoids the overhead associated with connections, error checks and the retransmission of missing data, it's suitable for real-time or high.

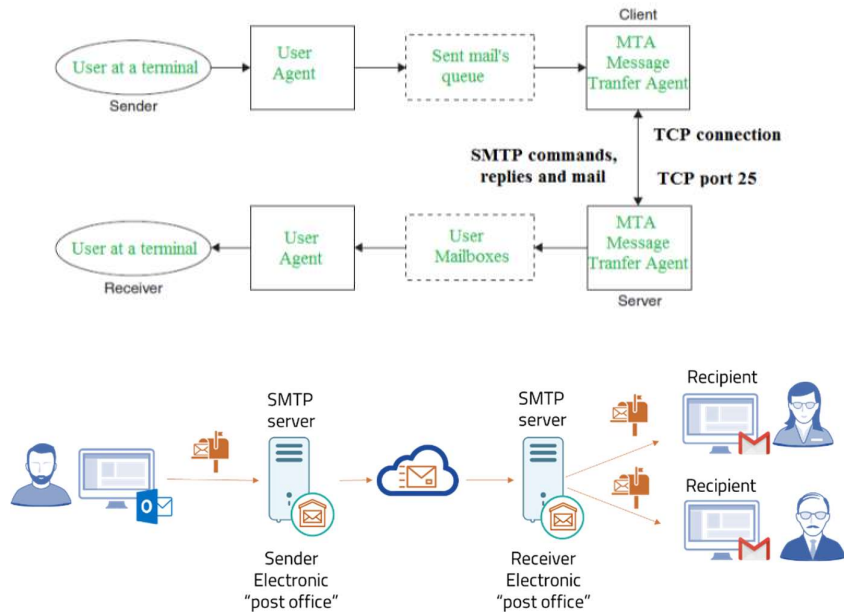
Advantages:

- **No retransmission delays** – UDP is suitable for time-sensitive applications that can't afford retransmission delays for dropped packets. Examples include Voice over IP (VoIP), online games, and media streaming.
- **Speed** – UDP's speed makes it useful for query-response protocols such as DNS, in which data packets are small and transactional.
- **Suitable for broadcasts** – UDP's lack of end-to-end communication makes it suitable for broadcasts, in which transmitted data packets are addressed as receivable by all devices on the internet. UDP broadcasts can be received by large numbers of clients without server-side overhead.

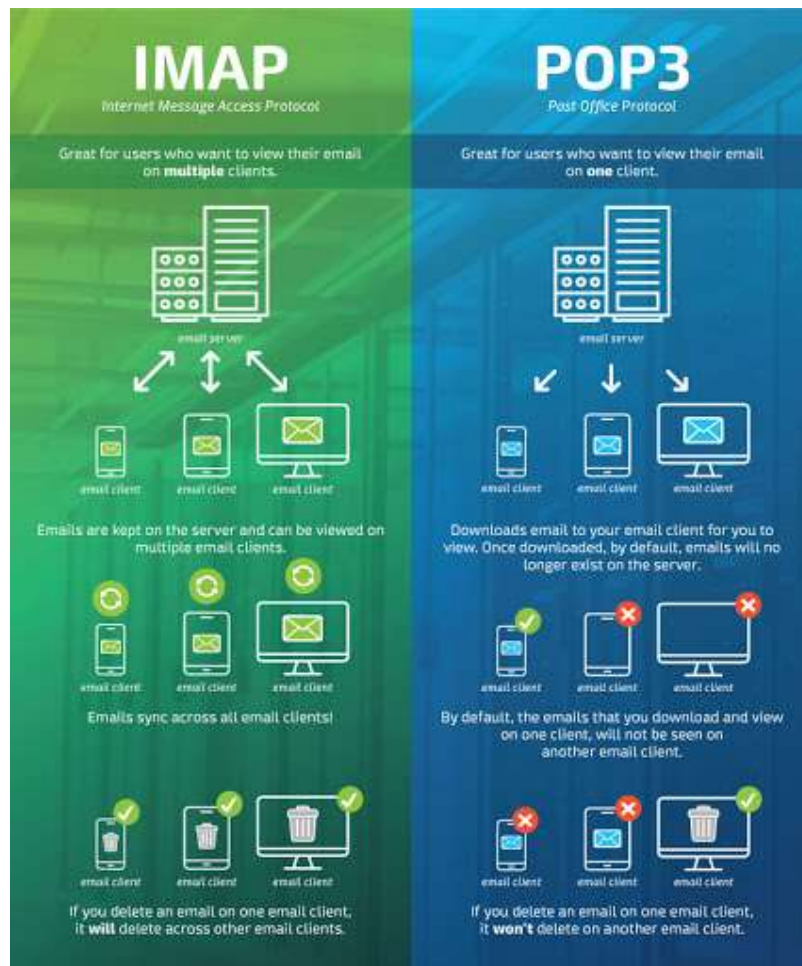
Disadvantages:

1. No guaranteed ordering of packets.
2. No verification of the readiness of the computer receiving the message.

3. No protection against duplicate packets.
2. No guarantee the destination will receive all transmitted bytes. UDP, however, does provide a checksum to verify individual packet integrity.
4. **Simple mail transport Protocol (SMTP): Port Number: 25**
SMTP is designed to send and distribute outgoing E-Mail.



5. **Post office Protocol (POP): Port Number: 110**
POP3 is designed for receiving incoming E-mails.
6. **IMAP4 (Internet Message Access Protocol): Port Number: 143**
 - IMAP allows you to access your email wherever you are, from any device.
 - When you read an email message using IMAP, you aren't actually downloading or storing it on your computer; instead, you're reading it from the email service.
 - As a result, you can check your email from different devices, anywhere in the world: your phone, a computer, a friend's computer.
 - IMAP only downloads a message when you click on it, and attachments aren't automatically downloaded. This way you're able to check your messages a lot more quickly than POP.



6. **File Transfer Protocol (FTP):** Port Number: 20,21

FTP allows users to transfer files from one machine to another. Types of files may include program files, multimedia files, text files, and documents, etc.

7. **Trivial File Transfer Protocol (TFTP):** Port Number: 69

Allows a client to get a file from or put a file onto a remote host.

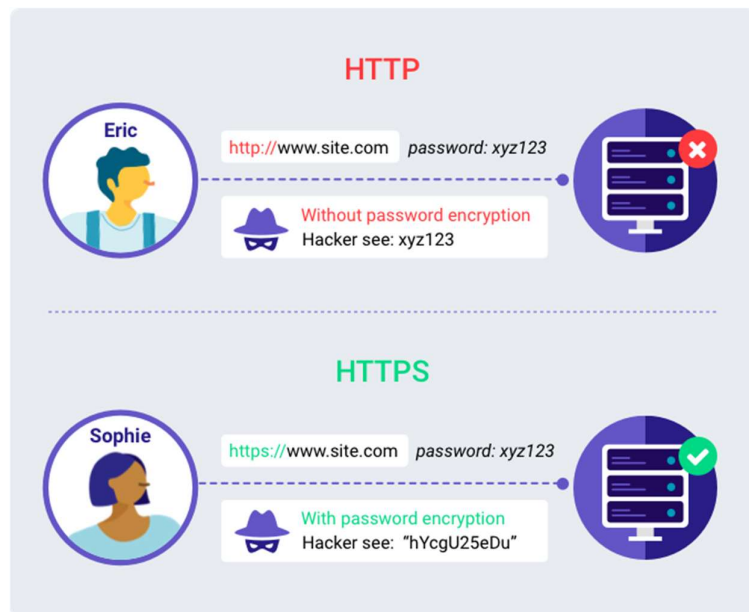
- Transferring files
- Remote-booting without hard drives
- Upgrading codes
- Backing up network configurations
- Backing up router configuration files
- Saving IOS images
- Booting PCs without a disk

FTP vs. TFTP

FTP	TFTP
2 connections: control (21) and data (20)	1 connection (69), stop and wait flow
Reliable service using TCP	Uses UDP, handles own retransmissions
Many commands	5 message types, only reads/writes files
Minimal security using logon procedure	No logon or security
Larger code size, full-featured	Lightweight, designed to fit on ROM

3. Hyper Text Transfer Protocol (HTTP): Port Number: 80

- HTTP is designed for transferring a hypertext among two or more systems. HTML tags are used for creating links.
- These links may be in any form like text or images.
- HTTP is designed on Client-server principles which allow a client system for establishing a connection with the server machine for making a request.
- The server acknowledges the request initiated by the client and responds accordingly.

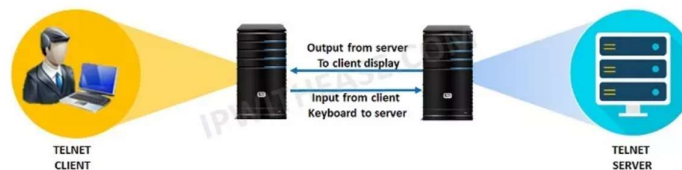


4. Hyper Text Transfer Protocol Secure (HTTPS): Port Number: 443

- HTTPS is abbreviated as Hyper Text Transfer Protocol Secure is a standard protocol to secure the communication among two computers one using the browser and other fetching data from web server.
- HTTP is used for transferring data between the client browser (request) and the web server (response) in the hypertext format, same in case of HTTPS except that the transferring of data is done in an encrypted format.
- So it can be said that https thwart hackers from interpretation or modification of data throughout the transfer of packets.

5. **Telnet:** Port Number: 23

Telnet is a set of rules designed for connecting one system with another. The connecting process here is termed as remote login. The system which requests for connection is the local computer, and the system which accepts the connection is the remote computer.

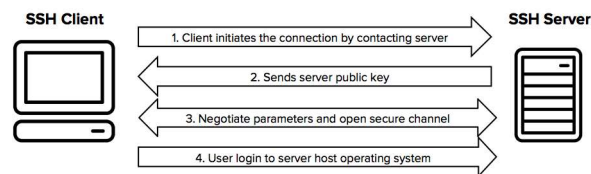
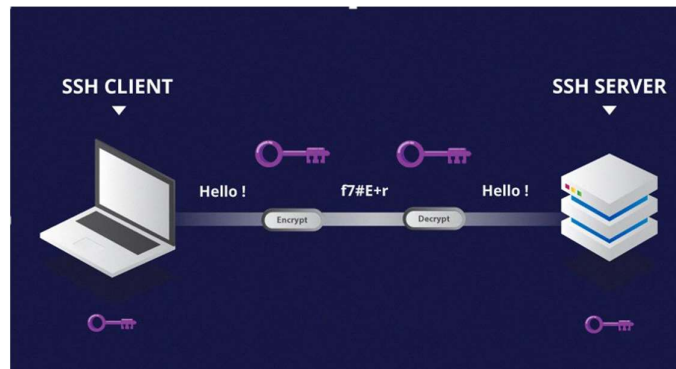


Telnet VS SSH

SSH(Secure Shell)	Telnet
Runs on port 22	Runs on port 23
Very Secure Protocol	Not Secure Protocol
Only major protocol to access	Joint abbreviation
Difficult to decrypt	No data encryption
All popular Operating System	Linux , Windows

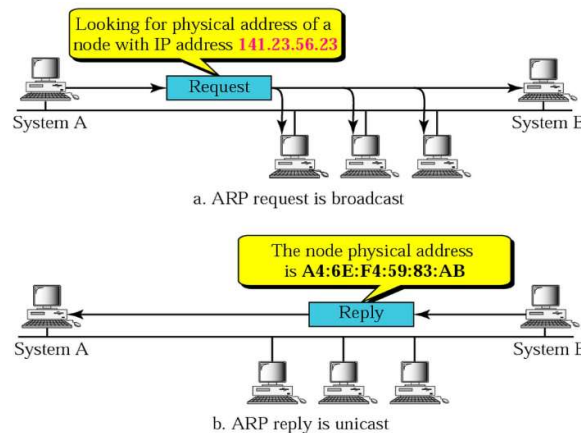
6. **SSH (Secure Shell):** Port Number: 22

- ✓ The SSH protocol (also referred to as Secure Shell) is a method for secure remote login from one computer to another.
- ✓ It provides several alternative options for strong authentication, and it protects the communications security and integrity with strong encryption.



7. ARP (Address Resolution Protocol):

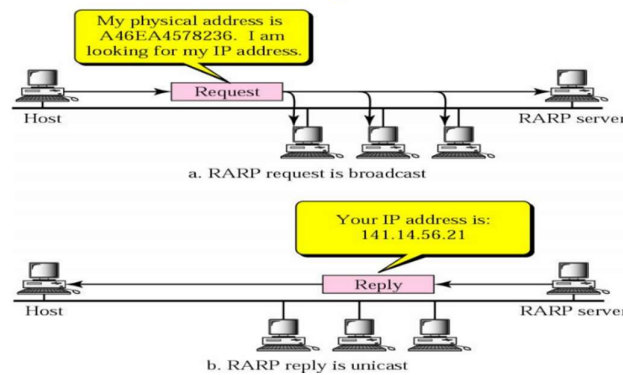
ARP finds the hardware address, also known as Media Access Control (MAC) address, of a host from its known IP address.



8. Reverse Address Resolution Protocol (RARP):

- ✓ **RARP** is an abbreviation for Reverse Address Resolution Protocol. **RARP** is a **TCP/IP** protocol that is responsible for the translation of Physical Address (e.g. – Ethernet address) to be translated into an **IP address**.
- ✓ Hosts like diskless workstations only have their hardware interface addresses or **MAC address**, but not their IP addresses. They must discover their **IP addresses** from an external source, usually via **RARP** protocol.

RARP Operation

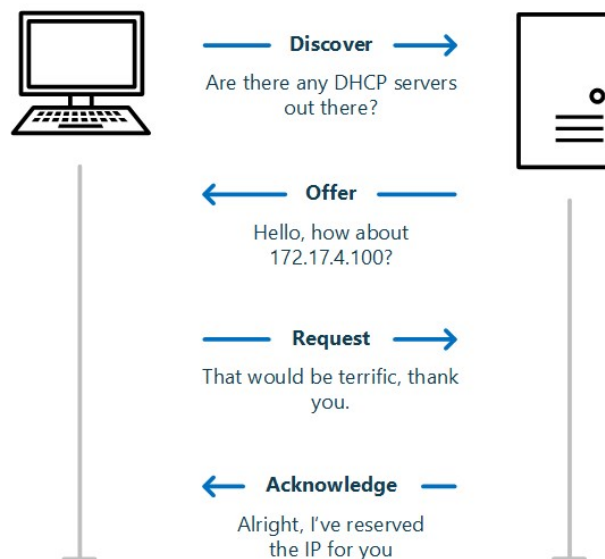


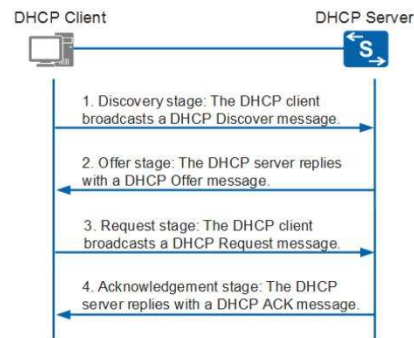
9. DHCP (Dynamic Host Configuration Protocol): Port Number: 67, 68

UDP port number 67 is used for the server and UDP port number 68 is used for the client.

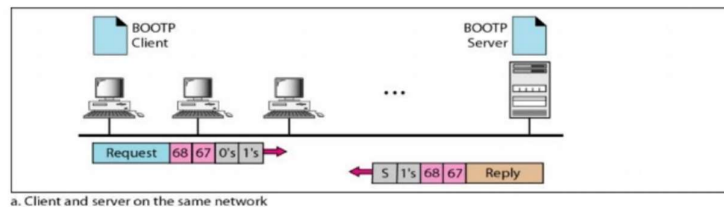
- Dynamic Host Configuration Protocol (DHCP) is a network protocol that is used to configure network devices to communicate on an IP network.
- A DHCP client uses the DHCP protocol to acquire configuration information, such as an IP address, a default route, and one or more DNS server addresses from a DHCP server.
- There are 4 types of DHCP packets:

- ✓ **DHCPDISCOVER**
- ✓ **DHCPOFFER**
- ✓ **DHCPREQUEST**
- ✓ **DHCPACK**





10. Bootstrap Protocol (BOOTP)



Bootstrap Protocol (BOOTP) is a networking protocol which is used by networking administration to give IP addresses to each member of that network for participating with other networking devices by the main server.

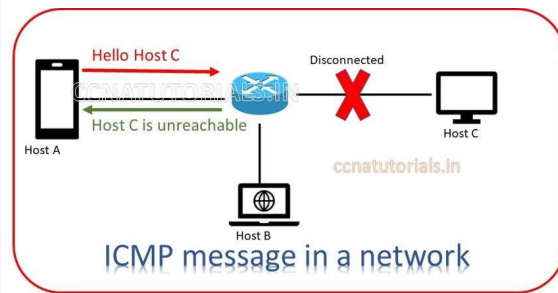
11. SNMP (Simple Network Management Protocol) *Port Number: Agents use UDP 161, while the manager uses UDP 162.*

- Simple Network Management Protocol (SNMP) is an Internet Standard protocol used to monitor and manage the network devices connected over an IP. Different devices like routers, switches, firewalls, load balancers, servers, CCTV cameras, and wireless devices communicate using SNMP.
- SNMP collects data from these devices, organizes them, and sends them for network monitoring and management with fault detection and isolation.

12. ICMP (ICMP Internet Control Message Protocol):

- ICMP Internet Control Message Protocol is an internet layer protocol in TCP/IP model.
- The ICMP protocol provides error reporting, troubleshooting of network devices in a network.
- ICMP works with IP protocol, as IP protocol is a connectionless and unreliable protocol.

- **ICMP is mostly used by ping and traceroute command.** Data lost reporting is main responsibility of ICMP protocol.
- Ping command show the data transmitted and received quantity.
- Counting of lost packets monitored and reported by the ICMP protocol to all device in a network.



13. Ping (Packet Internet or Inter-Network Groper):

- Is a computer network administration software utility used to test the reachability of a host on an Internet Protocol (IP) network.
- It is available for virtually all operating systems that have networking capability, including most embedded network administration software.

```
C:\>ping /?

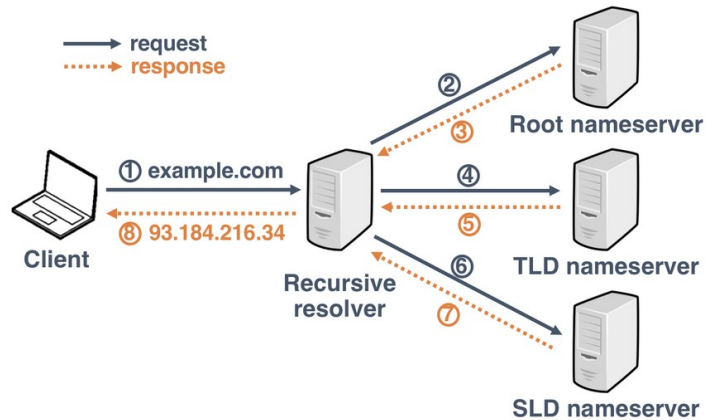
Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
          [-r count] [-s count] [[-j host-list] | [-k host-list]]
          [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
          [-4] [-6] target_name

Options:
  -t          Ping the specified host until stopped.
              To see statistics and continue - type Control-Break;
              To stop - type Control-C.
  -a          Resolve addresses to hostnames.
  -n count    Number of echo requests to send.
  -l size     Send buffer size.
  -f          Set Don't Fragment flag in packet (IPv4-only).
  -i TTL      Time To Live.
  -v TOS      Type Of Service (IPv4-only. This setting has been deprecated
              and has no effect on the type of service field in the IP
```

14. DNS Domain Name System: Port Number-53

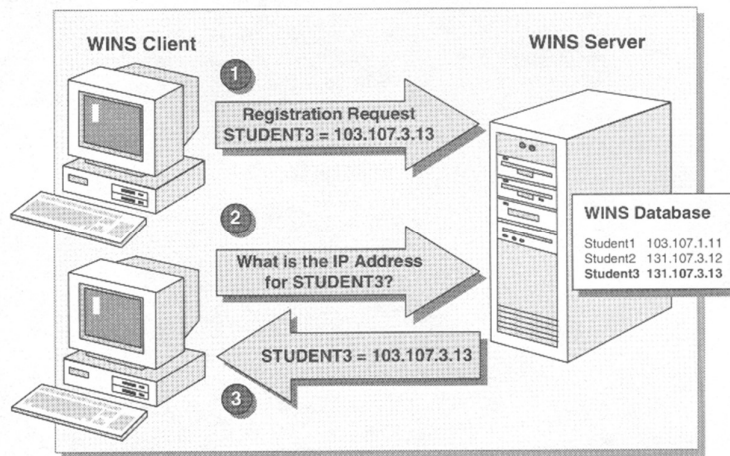
- The Domain Name System (DNS) turns domain names into IP addresses, which browsers use to load internet pages.
- Every device connected to the internet has its own IP address, which is used by other devices to locate the device.

- DNS servers make it possible for people to input normal words into their browsers, such as Fortinet.com, without having to keep track of the IP address for every website.



15. WINS Windows Internet Name Services:

Windows Internet Name Service (WINS) is the Microsoft implementation of NetBIOS Name Service (NBNS), a name server and service for NetBIOS computer names. Effectively, WINS is to NetBIOS names what DNS is to domain names — a central mapping of host names to network addresses.



12. SIP (Session Initiation Protocol): Port Number: 5060 or 5061

13. RTP (Real-Time Transport Protocol)

